2018B F=ma Exam: Problem 11

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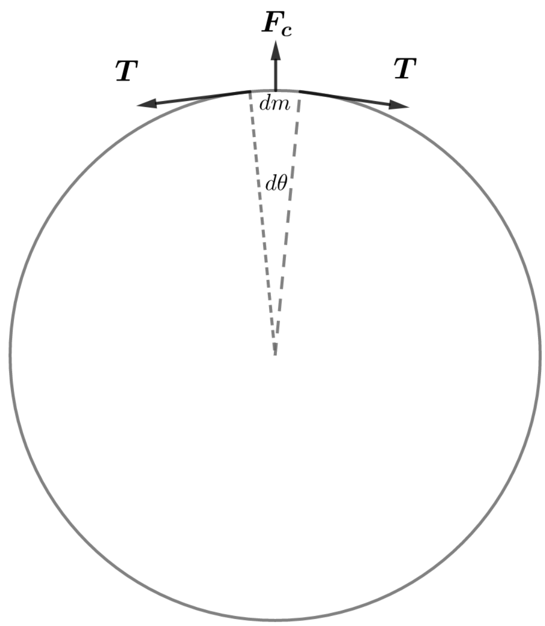

In the rotating frame, we have

$$2T\sin\left(\frac{d\theta}{2}\right) = F_c = dm\,\omega^2 R$$

Keeping to first order,

$$2T\left(\frac{d\theta}{2}\right) = dm\,\omega^2 R$$

$$T = \frac{dm}{d\theta}\,\omega^2 R = \frac{M\omega^2 R}{2\pi}$$

Recall the speed of waves in a string is given by

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{M/L}}$$

Thus, we have

$$v = \sqrt{\frac{M\omega^2 R}{2\pi}\frac{2\pi R}{M}} = \omega R$$

If the angular velocity is doubled, then the speed of transverse waves is also doubled so the answer is $\boxed{\text{C}}$.